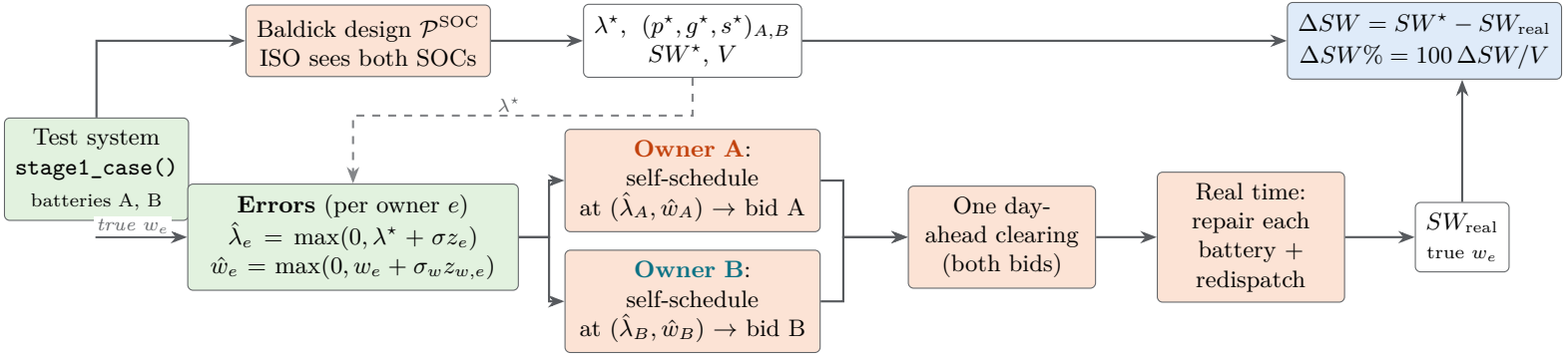

Baldick design $\mathcal{P}^{\text{SOC}}$
ISO sees both SOCs
$\lambda^\star,\ (p^\star, g^\star, s^\star)_{A,B}$
$SW^\star,\ V$
$\Delta SW = SW^\star - SW_{\text{real}}$
$\Delta SW\% = 100\,\Delta SW/V$
$\lambda^\star$
Test system
stage1_case()
batteries A, B
true w_e
Errors (per owner e)
$\hat{\lambda}_e = \max(0, \lambda^\star + \sigma z_e)$
$\hat{w}_e = \max(0, w_e + \sigma_w z_{w,e})$
Owner A:
self-schedule
at $(\hat{\lambda}_A, \hat{w}_A) \to$ bid A
Owner B:
self-schedule
at $(\hat{\lambda}_B, \hat{w}_B) \to$ bid B
One day-ahead clearing (both bids)
Real time: repair each battery + redispatch
SW_{real}
true w_e